

Day 40 - Write code (5 Times)

1.

#data collection

```
df=pd.read_csv("iris.csv")
```

```
df1=pd.read_excel("customers.xlsx")
```

```
df2=sns.load_dataset("titanic")
```

```
from sklearn.datasets import load_iris
```

```
iris=load_iris()
```

```
x=iris.data
```

```
y=iris.target
```

#data preparation

```
df.head()
```

```
df.tail()
```

```
df.info()
```

```
df.shape
```

```
df.size
```

```
df.index
```

```
df.describe()
```

#data cleaning

```
df.isnull().sum()
```

#setup libraries:

```
import numpy as np
```

```
import pandas as pd
```

```
from sklearn.model_selection import
```

```
train_test_split
```

#train_test_split

```
x_train,x_test,y_train,y_test=train_test_split(x  
y,test_size=0.2,random_state=42)
```

#regression

```
from sklearn.linear_model import
```

```
LinearRegression
```

```
model=LinearRegression()
```

```
model.fit(x_train,y_train)
```

```
y_pred=model.predict(x_test)
```

```
from sklearn.linear_model import Ridge
```

```
model=Ridge(alpha=1.0)
```

```
model.fit(x_train,y_train)
```

```
from sklearn.linear_model import Lasso
```

```
model=Lasso(alpha=0.1)
```

```
model.fit(x_train,y_train)
```

```
from sklearn.preprocessing import  
PolynomialFeatures  
model=PolynomialFeatures(degree=3)  
model.fit_transform(x)
```

```
#classification  
from sklearn.linear_model import  
LogisticRegression  
model=LogisticRegression()  
model.fit(x_train,y_train)
```

```
from sklearn.neighbors import  
KNeighborsClassifier  
model=KNeighborsClassifier(n_neighbors=5)  
model.fit(x_train,y_train)
```

```
from sklearn.svm import SVC  
model=SVC(kernel='rbf')  
model.fit(x_train,y_train)
```

```
from sklearn.tree import DecisionTreeClassifier  
model=DecisionTreeClassifier()  
model.fit(x_train,y_train)
```

```
from sklearn.ensemble import  
RandomForestClassifier  
model=RandomForestClassifier()  
model.fit(x_train,y_train)
```

```
from sklearn.naive_bayes import GaussianNB  
model=GaussianNB()  
model.fit(x_train,y_train)
```

```
#clustering  
from sklearn.cluster import KMeans  
model=KMeans(n_clusters=3)  
model.fit(x)  
labels=model.predict(x)
```

```
from sklearn.cluster import DBSCAN  
model=DBSCAN(eps=0.5,min_samples=5)  
labels=model.fit_predict(x)
```

```
from sklearn.mixture import GaussianMixture  
model=GaussianMixture(n_components=3)  
model.fit(x)  
labels=model.predict(x)
```



```
from sklearn.cluster import  
AgglomerativeClustering  
model=AgglomerativeClustering(n_clusters=2)  
labels=model.fit_predict(x)
```

```
from mlxtend.frequent_patterns import  
apriori, association_rules  
from mlxtend.frequent_patterns import  
fpgrowth, association_rules  
2.
```

```
#data collection  
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```
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Agglomerativeclustering
model=Agglomerativeclustering(n_clusters=2)
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```
from mlxtend.frequent_patterns import
apriori,association_rules
from mlxtend.frequent_patterns import
fpgrowth,association_rules
```

3.

```
#data collection
df=pd.read_csv("iris.csv")
df1=pd.read_excel("customers.xlsx")
```

```
df2=sns.load_dataset("titanic")
```

```
from sklearn.datasets import load_iris
```

```
iris=load_iris()
```

```
x=iris.data
```

```
y=iris.target
```

```
#data preparation
```

```
df.head()
```

```
df.tail()
```

```
df.info()
```

```
df.shape
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df.size
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df.describe()
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#data cleaning
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```
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```
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```
#regression
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```
LinearRegression
```

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model=LinearRegression()
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model.fit(x_train,y_train)
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y_pred=model.predict(x_test)
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```
from sklearn.linear_model import Lasso
```

```
model=Lasso(alpha=0.1)
```

```
model.fit(x_train,y_train)
```

```
from sklearn.preprocessing import
```

```
PolynomialFeatures
```

```
model=PolynomialFeatures(degree=3)
```

```
model.fit_transform(x)
```

```
#classification
```

```
from sklearn.linear_model import
```

```
LogisticRegression
```

```
model=LogisticRegression()
```

```
model.fit(x_train,y_train)
```

```
from sklearn.neighbors import
```

```
KNeighborsClassifier
```

```
model=KNeighborsClassifier(n_neighbors=5)
```

```
model.fit(x_train,y_train)
```

```
from sklearn.svm import SVC
```

```
model=SVC(kernel='rbf')
```

```
model.fit(x_train,y_train)
```

```
from sklearn.tree import DecisionTreeClassifier
```

```
model=DecisionTreeClassifier()
```

```
model.fit(x_train,y_train)
```

```
from sklearn.ensemble import
```

```
RandomForestClassifier
```

```
model=RandomForestClassifier()
```

```
model.fit(x_train,y_train)
```

```
from sklearn.naive_bayes import GaussianNB  
model=GaussianNB()  
model.fit(x_train,y_train)
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from sklearn.cluster import KMeans  
model=KMeans(n_clusters=3)  
model.fit(x)  
labels=model.predict(x)
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from sklearn.cluster import DBSCAN  
model=DBSCAN(eps=0.5,min_samples=5)  
labels=model.fit_predict(x)
```

```
from sklearn.mixture import GaussianMixture  
model=GaussianMixture(n_components=3)  
model.fit(x)  
labels=model.predict(x)
```

```
from sklearn.cluster import  
Agglomerativeclustering  
model=Agglomerativeclustering(n_clusters=2)  
labels=model.fit_predict(x)
```

```
from mlxtend.frequent_patterns import  
apriori, association_rules  
from mlxtend.frequent_patterns import  
fpgrowth, association_rules
```

4.

```
#data collection  
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df1=pd.read_excel("customers.xlsx")  
df2=sns.load_dataset("titanic")
```

```
from sklearn.datasets import load_iris  
iris=load_iris()  
x=iris.data  
y=iris.target
```

```
#data preparation  
df.head()  
df.tail()  
df.info()  
df.shape  
df.size  
df.index
```

```
df.describe()
```

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df.isnull().sum()
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#setup libraries:
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from sklearn.model_selection import  
train_test_split
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```
#train_test_split
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x_train,x_test,y_train,y_test=train_test_split(x  
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```
from sklearn.linear_model import Ridge
```

```
model=Ridge(alpha=1.0)
```

```
model.fit(x_train,y_train)
```

```
from sklearn.linear_model import Lasso  
model=Lasso(alpha=0.1)  
model.fit(x_train,y_train)
```

```
from sklearn.preprocessing import  
PolynomialFeatures  
model=PolynomialFeatures(degree=3)  
model.fit_transform(x)
```

```
#classification
```

```
from sklearn.linear_model import  
LogisticRegression  
model=LogisticRegression()  
model.fit(x_train,y_train)
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Agglomerativeclustering
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```
from mlxtend.frequent_patterns import
apriori,association_rules
from mlxtend.frequent_patterns import
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```

5.

```
#data collection
df=pd.read_csv("iris.csv")
df1=pd.read_excel("customers.xlsx")
df2=sns.load_dataset("titanic")
```

```
from sklearn.datasets import load_iris
iris=load_iris()
```

```
x=iris.data
```

```
y=iris.target
```

```
#data preparation
```

```
df.head()
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#data cleaning
```

```
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```

```
model.fit(x_train,y_train)
```

```
from sklearn.linear_model import Lasso
```

```
model=Lasso(alpha=0.1)
```

```
model.fit(x_train,y_train)
```

```
from sklearn.preprocessing import
```

```
PolynomialFeatures
```

```
model=PolynomialFeatures(degree=3)
```

```
model.fit_transform(x)
```

#classification

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from sklearn.linear_model import
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LogisticRegression
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model=LogisticRegression()  
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```
#clustering
```

```
from sklearn.cluster import KMeans
```

```
model=KMeans(n_clusters=3)
```

```
model.fit(x)
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```
labels=model.predict(x)
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from sklearn.cluster import DBSCAN
```

```
model=DBSCAN(eps=0.5,min_samples=5)
```

```
labels=model.fit_predict(x)
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```
from sklearn.mixture import GaussianMixture
```

```
model=GaussianMixture(n_components=3)
```

```
model.fit(x)
```

```
labels=model.predict(x)
```

```
from sklearn.cluster import
```

```
Agglomerativeclustering
```

```
model=Agglomerativeclustering(n_clusters=2)
```

```
labels=model.fit_predict(x)
```

```
from mlxtend.frequent_patterns import
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apriori,association_rules
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fpgrowth, association_rules